

Week 4 — If/Else Decisions

Name: ____ Date: ____

A comparison is a **question**. The computer answers 1 (yes) or 0 (no). *if* runs its block only when the answer is 1 — and *end* closes the decision.

Today's words

Word	What it means
logical	the computer's answer to a question: 1 = yes, 0 = no
comparison	a question about values: > < >= <= == ~=
condition	the question an <i>if</i> checks before deciding
if / elseif / else	if this... otherwise if this... otherwise. ONE branch runs
end	closes the decision — every <i>if</i> gets one
~=	not equal (the ~ is a wiggly NOT)

1 • Be the computer 🤖

Trace top-down and write the ONE line that prints:

```
temperature = 12;
if temperature >= 28
    disp('melting')
elseif temperature >= 15
    disp('perfect')
elseif temperature >= 0
    disp('jacket')
else
    disp('frozen')
end
```

The screen shows: ____

2 • Closed or open circle? ◦ •

The ride rule is height >= 132 . Circle the bouncer's answer for each rider:

Rider's height	Rides?
131	RIDES / TEACUPS
132	RIDES / TEACUPS
133	RIDES / TEACUPS

On a number line, >= 132 is a **CLOSED / OPEN** (circle one) circle at 132. If the rule were > 132 , the rider at exactly 132 would: ____

3 • Spot the bug 🐛

Each snippet has **one** mistake. Circle it.

```
1. if age = 13
    disp('teen')
end

2. if age >= 13
    disp('teen')

3. if age != 13
    disp('not thirteen')
end
```

🧠 Brain teaser (optional — take it home)

The hat with a hole in it. A student wrote this Sorting Hat, but ONE lucky number makes it misbehave. Which number, and what happens to it?

```
if lucky > 50
    disp('Team Waffle')
elseif lucky < 50
    disp('Team Goat')
end
```

Then fix it: rewrite the rules (you're allowed to change > and < to anything from today) so that EVERY number from 1 to 100 lands on exactly one team. Bring your fix next week for a shout-out — bonus glory if you find *two* different ways to plug the hole.